

On time.

$$\lim_{s \rightarrow 0} P(s) = 0, \quad \lim_{P \rightarrow 0} T(P) = 0$$

Q/S: On Saturday, Somewhere.

Abstract

Time cannot exist without two points.

Through mathematical derivation within the frameworks of Special and General Relativity, we demonstrate that time collapses in the absence of spatial differentiation, reinforcing its role as an emergent consequence rather than an absolute structure.

Furthermore, we assert that time is subordinate to the constraints of the limits dimensions (ω), and (ζ) and the probabilistic nature of chance (ξ), aligning with the 7dU framework's approach to fundamental reality. Extreme curvature scenarios, including gravitational singularities (*zero*, ζ) and cosmic boundary conditions (*infinity*, ω)—provide natural tests of this proposition, where time may dissolve or reconfigure based on the geometric properties of space itself.

This proof does not argue for time's existence—it simply exposes its dependence. Without space, there are no points. Without points, there is no time.

1. Introduction

Time is not an independent entity. It exists only in relation to spatial separation. If there is no distance between two points, there is no meaningful way to define time.

This proof establishes that time is a property of space, not a fundamental dimension. Using the formalism of Special and General Relativity, we show that time collapses when spatial differentiation vanishes. Time is governed by the constraints of the limits dimension (ω) and the probabilities of chance (ξ). It is not an absolute—it is an emergent effect.

2. Special Relativity: Time Requires Space

In Minkowski spacetime, the interval is defined as:

$$ds^2 = c^2 dt^2 - dx^2$$

Solving for time:

$$dt = \pm \frac{\sqrt{ds^2 + dx^2}}{c}$$

If $dx = 0$ (no spatial separation), the equation collapses. Time loses definition.

3. General Relativity: Time Depends on Geometry

In a curved spacetime, the interval is given by:

$$ds^2 = g_{tt}c^2 dt^2 - g_{xx}dx^2$$

Solving for time:

$$dt = \pm \frac{\sqrt{ds^2 + dx^2 g_{xx}(s)}}{c\sqrt{g_{tt}(s)}}$$

If $dx = 0$, time's definition depends entirely on the curvature of space. In extreme cases, such as singularities or asymptotic infinities, time ceases to function as a meaningful parameter.

$$dt = \pm \frac{\sqrt{ds^2 + dx^2 g_{xx}(s)}}{c\sqrt{g_{tt}(s)}}$$

If $dx = 0$, time is only what space allows it to be.

4. Logical Implications

If time were fundamental, it would exist without reference.
But the math does not allow it.

Without separation, there is no time.
Without space, there is no separation.

Time is not absolute.
It is the measure of difference.

5. Conclusion

$$dt = \pm \frac{\sqrt{ds^2 + dx^2 g_{xx}(s)}}{c\sqrt{g_{tt}(s)}}$$

Time is only what space allows it to be.